SENSEI / AKIRA

Product & Integration Briefing

The Autonomous Desktop AI Operator — Native Integrations, Real-Time Voice & Safety Architecture

CONFIDENTIAL — For Investors & Partners
Platform: Windows 10/11 Native | Status: V1 Production Live
Date: September 2026

⚡ Product Vision: Direct Native OS Automation & Real-Time Voice

Sensei is an **autonomous desktop AI operator** built to execute real knowledge work directly on the user's computer. Instead of confining users to a browser chat window, Sensei lives inside the operating system—controlling Microsoft Office via native Win32 COM, maintaining personal knowledge graphs via Obsidian, driving isolated web sessions via Selenium, and listening with zero-latency local voice recognition. It bridges high-level intelligence with deterministic desktop execution.

< 0.1 ms	< 50 ms	< 1.0 ms	67 Modules
PATTERN PRE-ROUTER	VOICE BARGE-IN	STATIC SAFETY GUARD	LIVE CODEBASE

1. WHAT SENSEI DOES RIGHT NOW (V1 PRODUCTION CORE)

📊 1. Native Win32 COM Office Automation

Direct programmatic control of desktop productivity suites with 100% deterministic precision:

- **Microsoft Excel:** Cell-by-cell manipulation, dynamic formula writing (`=SUM`, `=VLOOKUP`, `=IRR`), automated cell formatting/color palettes, and live chart generation.
- **Microsoft Word:** Automated document creation, heading styling, table inserts, and executive summary exports.
- **Microsoft PowerPoint:** Full deck generation, slide layout creation, bullet formatting, and speaker notes.

🧠 2. Obsidian "Second Brain" Graph Integration

Deep bidirectional connection with the user's local Obsidian knowledge vault:

- **Local REST API Architecture:** Communicates over secure local endpoints with zero cloud data exposure.
- **Knowledge Graph Creation:** Generates interconnected markdown notes, bi-directional backlinks (`[[Topic]]`), and hierarchical tags.
- **Instant Context Retrieval:** Pulls historical meeting notes, project specs, and references into live agent context.

🎙️ 3. Sub-100ms Voice & Interruption Engine

Conversational fluidity that feels natural and instantaneous:

- **Offline Wake Word Detection:** On-device ONNX engine (`sensei.onnx`) triggers instantly with zero cloud round-trip.
- **Silero VAD Barge-In:** Detects user speech interruptions in <50ms, immediately cutting agent speech for natural dialogue.
- **Natural Speech Synthesis:** Smooth text-to-speech with pitch calibration and dynamic playback control.

🛡️ 4. Fail-Closed Dual-Layer Safety Guard

Enterprise-grade safety architecture preventing rogue actions:

- **Static Verification (<1ms):** `guard_rules.py` pre-validates action names, parameter arity, and destructive commands offline.
- **Fail-Closed Guarantee:** If network or validation drops, execution immediately halts and requests user confirmation.
- **Trust Streak Ledger (`reliability.py`):** Earned autonomy protocol that tracks supervised executions before auto-approving routine steps.

2. NATIVE OS TOOLS & WEB CAPABILITIES IN V1

🌐 Isolated Web Automation

Pre-warmed Selenium Chrome instances with isolated profile directories for form fills, research, and authenticated workflows.

🖼️ Visual & Media Engine

Integrated `image_flow.py` pipeline that searches, previews, and downloads verified images without downloading web junk.

⚙️ OS Hardware & Apps

Native Win32 system controls: volume, brightness, process killer, hardware resource diagnostics, and app launching.

Sensei V2: Dedicated High-Value Modes

Specialized Intelligence for Students & Business Operators with Deep Domain Workflows

🎯 The V2 Upgrade: Hyper-Specialized Operational Modes

Sensei V2 transforms the core OS operator into two purpose-built specialized modes: **Scholar Mode** (teaching any concept with adaptive difficulty, dynamic diagrams, and active testing) and **Operator Mode** (acting as an executive business co-founder handling idea reviews, financial spreadsheets, email triage, and SaaS MVPs).

MODE A: SENSEI SCHOLAR ("THE 10X ADAPTIVE SUPER-TUTOR")

📖 1. 5-Level Adaptive Feynman

Cuts 30-minute video lecture fluff into 90-second mastery:

- **Level 1 (ELI5):** Intuitive real-world analogies and simple mental models.
- **Level 2 (Undergrad):** Core equations, technical definitions, and mechanisms.
- **Level 3 (Exam/AP/GRE):** Deep heuristics, edge cases, and problem-solving drills.
- **Level 4 (Expert):** First-principles proofs, research papers, and trade-offs.
- **Level 5 (Socratic):** Interactive guiding questions that test reasoning.

📊 2. Dynamic Diagram Pipeline

Visual-first explanations generated in real-time:

- **Mermaid & Flowcharts:** Instantly creates system architectures, biological cycles, and logic trees.
- **Verified Image Fetching:** Pulls real schematics, anatomical illustrations, and historical maps via `image_flow.py`.
- **LaTeX Formula Rendering:** Crisp mathematical and chemical equation displays in native view.

📝 3. Active Recall & Obsidian

Long-term retention engine built into your vault:

- **Oral & Terminal Quizzing:** Rapid-fire diagnostic checks that test conceptual retention.
- **Obsidian Flashcards:** Compiles flashcard decks with spaced-repetition tags (`#review/hard`) into local vaults.
- **Weakness Ledger:** Tracks diagnostic gaps to revisit during future study sessions.

MODE B: SENSEI OPERATOR ("THE EXECUTIVE AI CO-FOUNDER")

💡 1. "Devil's Advocate" Reviewer

Brutal, experienced business critique & sparring:

- **Stress-Testing Ideas:** Challenges CAC/LTV, unit economics, market size (TAM), and churn vulnerabilities.
- **Competitive Moat Analysis:** Identifies why competitors can replicate and formulates defensible moats.
- **Strategic Frameworks:** Instant generation of Lean Canvas, SWOT matrices, and Porter's Five Forces.

📈 2. Automated Excel Financials

Zero-effort corporate spreadsheet modeling:

- **Full Financial Models:** Constructs 3-year cash flow projections, P&L sheets, and break-even tables in Excel.
- **Live Dynamic Formulas:** Injects formulas (`=IRR`, `=NPV`), header styling, and growth charts.
- **Sensitivity Scenarios:** Generates multi-variable sensitivity tables (Bull, Base, Bear cases) in seconds.

✉️ 3. Safe Email & SaaS Builder

Executive workflow & rapid product scaffolding:

- **Email Triage & Digest:** Reads incoming mail via `gmail_control.py`, prioritizes urgent client threads, and creates digests.
- **Guard-Gated Send:** Drafts on-brand replies; sends only with dual-guard confirmation or earned trust.
- **SaaS MVP Scaffolder:** Scaffolds starter landing pages, database schemas, and CRUD API endpoints.

EXAMPLE HIGH-IMPACT V2 WORKFLOWS

🎓 Student Workflow: "Teach Me Quantum Entanglement"

Sensei explains the core principle using the twin coin analogy in 60s, renders a Mermaid state diagram, creates 3 flashcards in Obsidian, and quizzes the user on Bell's inequality.

💼 Founder Workflow: "Build Financial Model for B2B SaaS"

Sensei interrogates pricing ($49/mo), calculates CAC payback period, opens Excel, populates a 36-month P&L with active formulas, and drafts an investor email summary.

SENSEI MATRIX

Self-Learning Engine, Tiers & Delivery Roadmap

Autonomous Personalization, Feature Matrix by Tier & Fast-Track Engineering Timeline

Architecture & Tier Specification
Delivery Timeline: 12 – 15 Working Days

1. AUTONOMOUS SELF-LEARNING & MEMORY ENGINE

💉 Continuous Passive User Profiling

Sensei gets smarter with every interaction without costly model retraining:

- **Background Memory Observer:** `memory_observer.py` passively extracts user facts, communication tone, and operational boundaries.
- **Learning Velocity Ledger:** Remembers what study topics you mastered quickly vs. concepts that required repeated review.
- **Business Entity Memory:** Stores your active companies, key clients, product pricing, and vendor details in local Markdown profiles.

🔒 Data Sovereignty & Earned Trust

A compounding user relationship built on trust and privacy:

- **Local Profile Storage:** All memory resides on the user's SSD in `memory/core_profile.md` and Obsidian vaults.
- **Trust Streak Mechanism (`reliability.py`):** High-impact actions earn autonomous permissions through repeated safe executions.
- **Deterministic Rollback:** Tracks state changes to ensure safe reversibility for local files and workflows.

2. COMPLETE 4-TIER FEATURE & CAPABILITY MATRIX

Capability / Feature	Free Tier ($0)	Standard Tier	Pro Tier	Beta Tier
Voice Wake & Barge-In (ONNX + VAD)	FULL	FULL	FULL	EXPERIMENTAL AVATARS
System Control (Volume, Brightness, Apps)	FULL	FULL	FULL	FULL
Obsidian Second Brain Integration	READ & CREATE	FULL GRAPH & TAGS	FULL + DEEP CONTEXT	FULL
Web Search & Image Engine	BASIC SEARCH	IMAGE FLOW + DOWNLOADER	FULL BROWSER AUTOMATION	FULL
Scholar Mode (Feynman & Quizzes)	BASIC EXPLANATIONS	5-LEVEL FEYNMAN + QUIZZES	FULL DIAGRAMS & FLASHCARDS	PREVIEW FEATURES
Win32 Office (Excel, Word, PPT)	DISABLED	WORD DRAFTING	FULL FORMULAS, CHARTS, PPT	FULL
Operator Mode (Business & Models)	DISABLED	IDEA REVIEW	FULL P&L, SAAS MVP, SWOT	PREVIEW FEATURES
Email Automation (Gmail Triage & Send)	DISABLED	READ & TRIAGE DIGEST	FULL DRAFT & GUARDED SEND	FULL
Advanced Multi-Step Reasoning Models	FAST GROQ/FLASH	FLASH + GROQ (HIGHER QUOTA)	CLAUDE 3.5 SONNET / 72B	EXPERIMENTAL LOCAL MODELS

3. ENGINEERING DELIVERY ROADMAP (ASAP SCHEDULE)

Sprint 1 (Days 1–5): MVP Core

PHASE 1 COMPLETE

- Global Mode Switcher (Scholar / Operator)
- 5-Level Adaptive Feynman Engine
- Mermaid Diagram Generation
- "Devil's Advocate" Idea Reviewer

Sprint 2 (Days 6–10): Deep Tools

PHASE 2

- Win32 Excel Financial & P&L Models
- Obsidian Spaced-Repetition Flashcards
- Email Triage & Guard-Protected Send
- Image Engine Diagram Visualizer

Sprint 3 (Days 11–15): Intelligence

PHASE 3

- Continuous Self-Learning Memory Observer
- SaaS MVP Scaffolder Boilerplates
- Tier Quota Enforcement & Key Management
- Full Production Packaging

🚀 Product Summary for Investors

Sensei combines **deterministic desktop execution** (Win32 Office, Obsidian, Selenium), **zero-latency real-time voice**, and **fail-closed safety guards** into an indispensable everyday tool. By expanding into Scholar and Operator modes, Sensei commands both viral bottom-up student adoption and high-ARPU business workflow automation.